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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Time: 03 Hours

Code: 100106 ***Subject: Basic Electrical Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) In source transformation, a voltage source in series with a resistor can be converted into:
- (i) A current source in parallel with a capacitor
- (ii) A current source in parallel with a resistor
- (iii) A voltage source in parallel with a resistor
- (iv) A current source in series with a resistor

(b) According to Kirchhoff's Voltage Law (KVL), the sum of voltages around a closed loop is:
- (i) Equal to the largest voltage in the loop
- (ii) Zero
- (iii) Equal to the source voltage
- (iv) Always negative

(c) Which of the following is a bilateral element?
- (i) Diode
- (ii) Transistor
- (iii) Resistor
- (iv) Battery

(d) Which of the following quantities in a magnetic circuit is analogous to current in an electric circuit?
- (i) Magnetic reluctance
- (ii) Magnetic flux
- (iii) Magnetomotive force (MMF)
- (iv) Permeability

(e) The power factor in an AC circuit is defined as the ratio of:
- (i) Active power to reactive power
- (ii) Apparent power to active power
- (iii) Active power to apparent power
- (iv) Apparent power to reactive power

(f) Which connection type has a neutral point?
- (i) Delta connection
- (ii) Star connection
- (iii) Both star and delta
- (iv) None of the above

(g) In a DC generator, the EMF is generated due to:
- (i) Stationary magnetic field
- (ii) Stationary armature
- (iii) Relative motion between magnetic field and conductor
- (iv) High resistance in the circuit

(h) Which of the following losses occurs in a transformer?
- (i) Copper loss
- (ii) Friction loss
- (iii) Windage loss
- (iv) Commutation loss

(i) In a series RLC circuit, resonance occurs when:
- (i) $X_L = X_C$
- (ii) $X_L > X_C$
- (iii) $X_L = R$
- (iv) $R = X_C$

(j) The main function of a distribution board is to:
- (i) Measure electrical power
- (ii) Connect transformers
- (iii) Distribute electrical power to circuits
- (iv) Store electrical energy

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Q.2 Solve this question : **[14]**

Explain Thevenin's and Norton's Theorems. How can one be derived from the other? Support your answer with a circuit diagram and step-by-step solution.

Q.3 Solve both questions : **[14]**

(a) State and explain Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) with an illustrative circuit and solve a simple numerical example.

(b) Describe the process of source transformation and provide an example where a voltage source is transformed into a current source and vice versa.

Q.4 Solve both questions : **[14]**

(a) Describe the concept of magnetic reluctance and derive the formula for reluctance in terms of length, area, and permeability.

(b) Explain the analogy between electric and magnetic circuits. Draw a table comparing the corresponding quantities.

Q.5 Solve both questions : **[14]**

(a) Define and compare average value, RMS value, form factor, and peak factor for sinusoidal waveforms. Give appropriate expressions and examples.

(b) Explain the concept of phasors. How are sinusoidally varying voltages and currents represented using phasors?

Q.6 Solve this question : **[14]**

Describe the difference between star and delta connections. Provide diagrams and explain voltage and current relations for both.

Q.7 Solve both questions : **[14]**

(a) Describe the construction and working of a single-phase transformer. Derive the EMF equation.

(b) Explain the types of DC machines and differentiate between a DC motor and a DC generator.

Q.8 Solve this question : **[14]**

Describe the different types of residential and commercial wiring systems with diagrams.

Q.9 Write short notes on any two of the following: **[14]**

(a) Maximum Power Transfer Theorem

(b) B-H Curve

(c) Power Factor improvement Methods

(d) Working of Three Phase Induction Motor

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